

Analytical Problem

Solving Activity

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COURSE: Computer Vision.

1. Gn:-

$$G_x = 24, \quad G_y = 18.$$

Soln:-
Gradient Magnitude.

$$\begin{aligned} G &= \sqrt{G_x^2 + G_y^2} \\ &= \sqrt{24^2 + 18^2} \\ &= \sqrt{576 + 324} = \sqrt{900} = 30. \end{aligned}$$

$$\text{Gradient Magnitude} = 30.$$

Gradient Direction.

$$\theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

$$\theta = \tan^{-1} \left(\frac{18}{24} \right)$$

$$\theta = \tan^{-1} (0.75) \approx 36.87.$$

$$\text{Gradient Direction} \approx 36.87.$$

2.
Qdn 5

Give Gradient Magnitudes.

Pixel	P_1	P_2	P_3	P_4	P_5
Magnitude	25	50	90	60	30

Since the gradient direction is horizontal each pixel is compared with its left and right neighbours.

$P_1 = 25 \rightarrow$ Boundary Pixel $\rightarrow$ Retained.

$P_2 = 50 \rightarrow$ less than $P_3 (90) \rightarrow$ Suppressed.

$P_3 = 90 \rightarrow$ Greater than $P_2 (50)$ and

$P_4 (60) \rightarrow$ Retained.

$P_4 = 60 \rightarrow$ Less than $P_3 (90) \rightarrow$ Suppressed.

$P_5 = 30 \rightarrow$ Boundary Pixel $\rightarrow$ Retained.

Result After NMS:

Pixel	P_1	P_2	P_3	P_4	P_5
NMS Result	25	0	90	0	30

Remaining edge Pixels : P_1 , P_3 and P_5 .

3. Double Thresholding.

Soln:-

Given:-

High Threshold = 100

Low Threshold = 50.

Rules:

Magnitude $\geq 100 \rightarrow$ Strong Edge.

$50 \leq \text{Magnitude} < 100 \rightarrow$ Weak Edge.

PIXEL	MAGNITUDE	CLASSIFICATION
P_1	135	Strong Edge.
P_2	90	Weak Edge.
P_3	45	Non-Edge
P_4	120	Strong Edge.
P_5	70	Weak Edge.
P_6	20	Non-Edge.

Answers:

Strong Edge : P_1, P_4 .

Weak Edge : P_2, P_5 .

Non-Edge : P_3, P_6

4.
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PIXEL	P_1	P_2	P_3	P_4	P_5	P_6	P_7
MAGNITUDE	180	95	100	115	65	8 25	10

Thresholds:

- High = 100
- Low = 50

1. Strong Edges.

Magnitude ≥ 100 : P_1, P_4, P_7 .

2. Weak Edges.

$50 \leq \text{Magnitude} < 100$: P_2, P_5 .

3. Non - Edges.

Magnitude < 50 : P_3, P_6 .

4. Final Edge Pixels after Hysteresis.

- P_2 is connected to strong edge P_1 , so P_2 is retained.
- P_5 is not connected to any strong edge, so P_5 is removed.

Final Edges : P_1, P_2, P_4, P_7 .

5. Hough Circle Transform:

$$r = \sqrt{(x-a)^2 + (y-b)^2}$$

i) Edge Points:

$$P_1 = (5, 4), \quad P_2 = (7, 6)$$

$$C_1 = (2, 4), \quad C_2 = (4, 3)$$

For $C_1 = (2, 4)$.

$$\begin{aligned} \bullet \quad P_1 \text{ and } C_1 : r &= \sqrt{(5-2)^2 + (4-4)^2} \\ &= \sqrt{3^2 + 0^2} = \sqrt{9} = 3 \end{aligned}$$

$$\boxed{r = 3}$$

$$\begin{aligned} \bullet \quad P_2 \text{ and } C_2 : r &= \sqrt{(7-2)^2 + (6-4)^2} \\ &= \sqrt{25+4} = \sqrt{29} = 5.39 \end{aligned}$$

$$\boxed{r = 5.39}$$

$$\begin{aligned} \bullet \quad P_1 \text{ and } C_2 : r &= \sqrt{(5-4)^2 + (4-3)^2} \\ &= \sqrt{1+1} = \sqrt{2} = 1.41 \end{aligned}$$

$$\boxed{r = 1.41}$$

$$\begin{aligned}
 P_2 \text{ and } C_2 : r &= \sqrt{(7-4)^2 + (6-3)^2} \\
 &= \sqrt{9+9} = \sqrt{18} = 4.24
 \end{aligned}$$

$$r = 4.24$$

6. Calculate the Radius:

Given:

$$r = (8, 6), C = (4, 3)$$

$$\begin{aligned}
 r &= \sqrt{(x-a)^2 + (y-b)^2} \\
 &= \sqrt{(8-4)^2 + (6-3)^2} = \sqrt{25} \quad \boxed{r = 5}
 \end{aligned}$$

7. Calculate the Radius:

$$C = (5, 4), P = (8, 8)$$

$$\begin{aligned}
 r &= \sqrt{(x-a)^2 + (y-b)^2} \\
 &= \sqrt{(8-5)^2 + (8-4)^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16} \\
 &= \sqrt{25} = 5
 \end{aligned}$$

$$r = 5$$